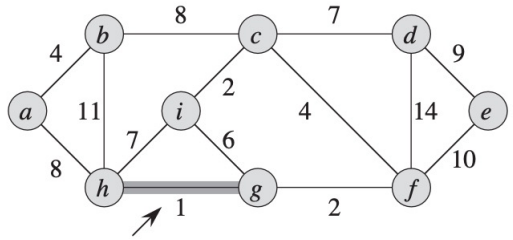
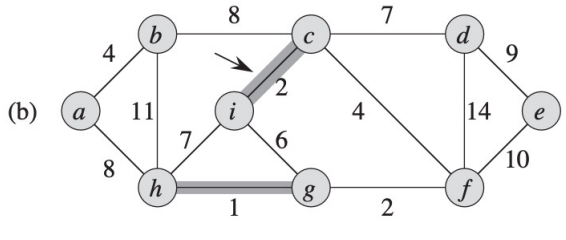
MSPs

```
MST-KRUSKAL(G, w)
1  A = ∅
2  for each vertex v ∈ G.V
3       MAKE-SET(v)
4  sort the edges of G.E into nondecreasing order by weight w
5  for each edge (u, v) ∈ G.E, taken in nondecreasing order by weight
6       if FIND-SET(u) ≠ FIND-SET(v)
7            A = A ∪ {(u, v)}
8            UNION(u, v)
9  return A
```

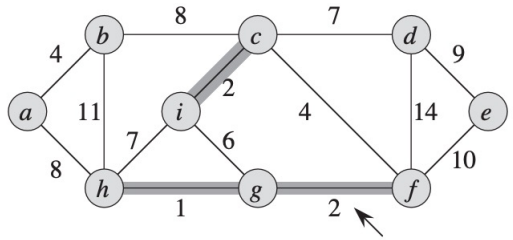
(a)



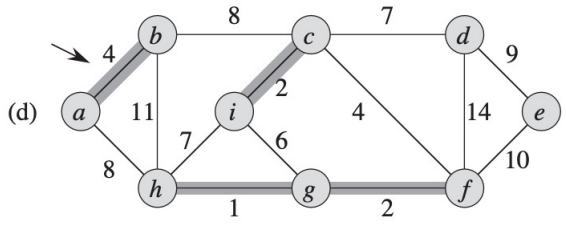
(b)



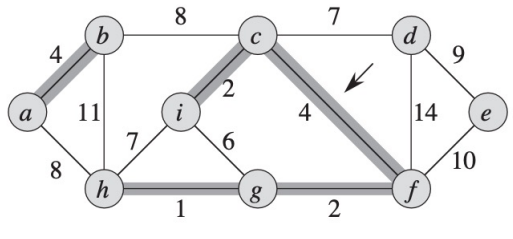
(c)



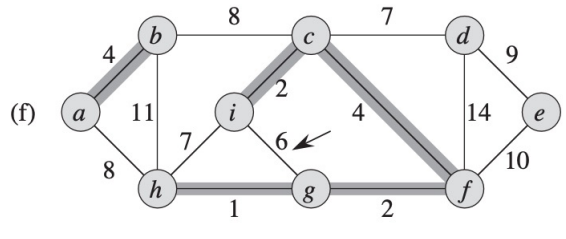
(d)



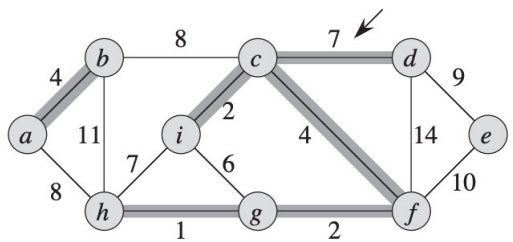
(e)



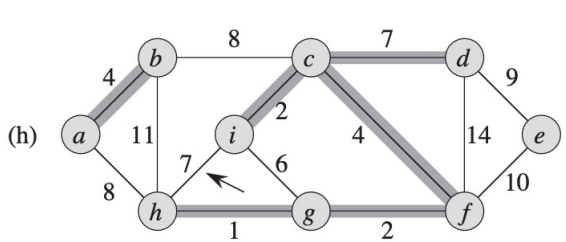
(f)



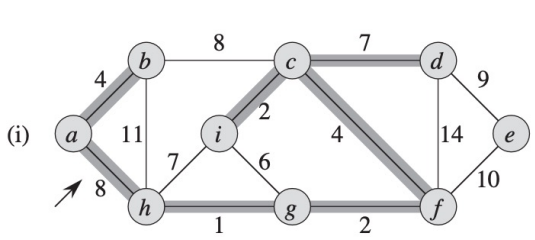
(g)



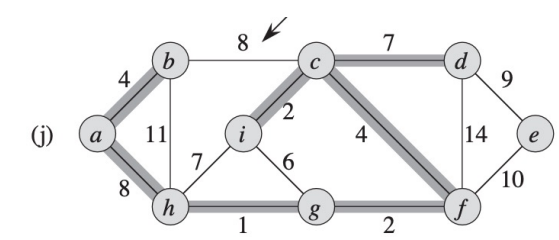
(h)



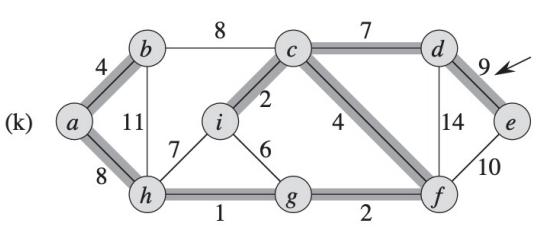
(i)



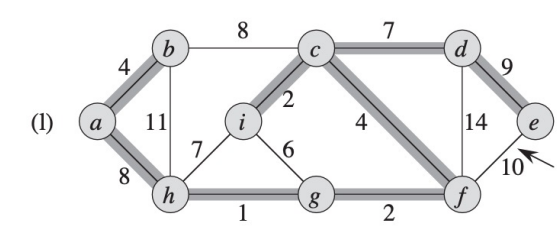
(j)



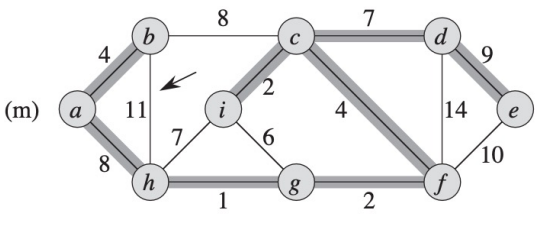
(k)



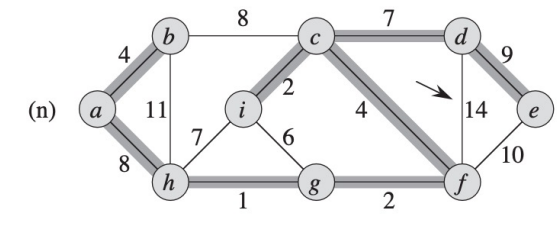
(l)



(m)

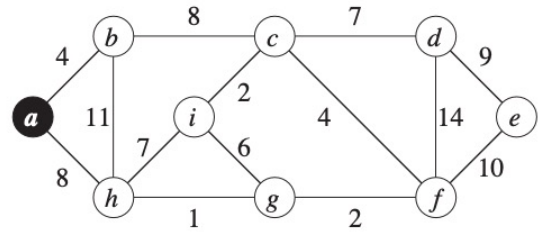


(n)

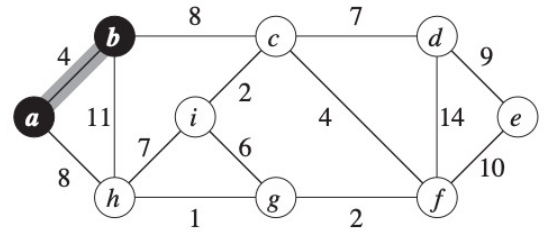


```
MST-PRIM(G, w, r)
 1  for each u ∈ G.V
 2      u.key = ∞
 3      u.π = NIL
 4  r.key = 0
 5  Q = G.V
 6  while Q ≠ ∅
 7      u = EXTRACT-MIN(Q)
 8      for each v ∈ G.Adj[u]
 9          if v ∈ Q and w(u, v) < v.key
10              v.π = u
11              v.key = w(u, v)
```

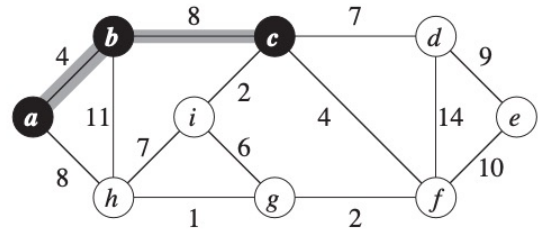
(a)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2

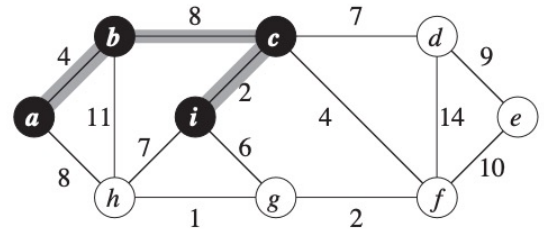
(b)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2

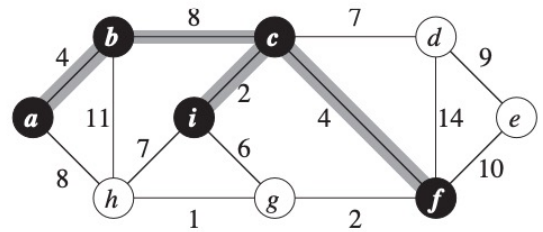
(c)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2

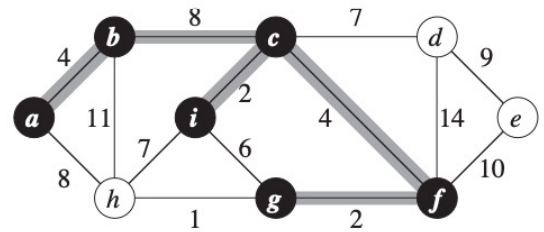
(d)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2

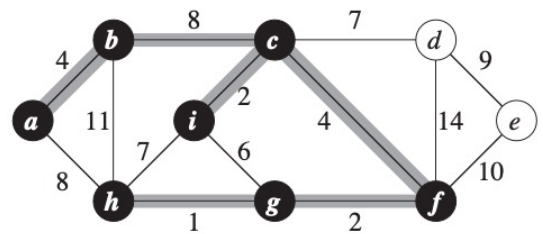
(e)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2

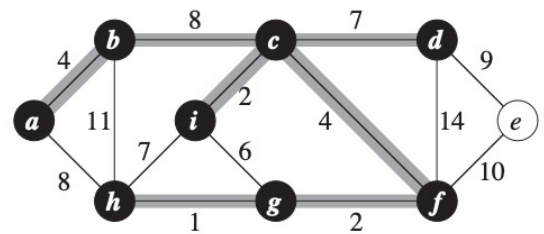
(f)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2

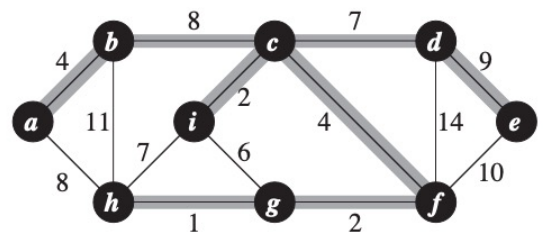
(g)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2

(h)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2

(i)


a
b
c
d
e
f
g
h
i
4
8
8
7
9
11
7
2
4
14
6
10
1
2